

# Piezoelectricity in the Proteinogenic Amino Acid L-leucine: A Novel Piezoactive Bioelectret

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## ABSTRACT

Here, we report the growth of bioelectret crystal films of L-leucine on conductive substrates and detail the first quantitative measurements of the direct piezoelectric effect in this proteinogenic amino acid. Through extensive electromechanical characterisation, we demonstrate that L-leucine is a promising candidate material for use in non-toxic, biocompatible and biodegradable energy harvesting and sensing devices. The data presented here is among, if not the, largest set of *quasi*-static, longitudinal piezoelectric measurements on crystalline films of a proteinogenic amino acid to date. We substantiate these measurements using a combination of density functional theory calculations and optical microscopy. Our data provides a further example of the enormous unexploited potential of amino acids and similar biological materials in flexible energy harvesting applications. Their combination of sizeable piezoelectric strain constants and extremely low elastic and dielectric constants makes them ideal as biocompatible replacements for toxic, expensive inorganic piezoelectric materials.

Index Terms — piezoelectricity, amino acid crystals, organic molecules, bioelectret phenomena, quantum mechanical simulation

## 1 INTRODUCTION

**PIEZOELECTRICITY** is an important functional property that allows certain materials to generate electricity when stressed or conversely, change shape when subjected to an electric field, and has been observed in a number of biological materials. The effect has been demonstrated in biological structures from bone [1] and fibrous proteins [2,3] to viruses [4], globular proteins [5] and their building blocks, the amino acids [6–10]. It has long been suggested that the key to understanding biological piezoelectricity is to start at the bottom of the biological hierarchy and study the 20 amino acids that make up proteins [11]. More quantitative insight into the piezoelectric effect in these compounds will accelerate their uptake as technologically viable alternatives to classical piezoelectrics that often contain toxic materials.

One challenge in obtaining a quantitative electromechanical response from materials such as amino acids is their inherent softness, which limits the use of conventional piezoelectric measurement techniques. Previous quantitative studies into amino acid piezoelectricity have utilised single crystals of glycine in both the gamma and beta crystalline phases [6], but it is very difficult to grow amino acid single crystals to sizes

large enough for use in technological applications. Thus, fabrication of devices that use piezoelectric amino acids require polycrystalline films that are robust enough to sustain multiple compressions. This has been accomplished using the non-proteinogenic racemic amino acid DL-alanine [10], and the purpose of the present study is to investigate the piezoelectric response in films of L-leucine, an amino acid directly incorporated in biological proteins.

L-leucine is an essential amino acid, falling into the class of proteinogenic amino acids that the body cannot synthesise. It is important for tissue growth and repair, stimulating myofibrillar muscle protein synthesis [12]. In the solid state, L-leucine crystallises in the monoclinic  $P2_1$  space group, which confers certain tensor properties including second harmonic generation and piezoelectricity. Here we present the first quantitative *quasi*-static piezoelectric measurements on crystal films of L-leucine. These measurements, coupled with quantum mechanical calculations, suggest L-leucine may be a suitable biocompatible, biodegradable replacement for traditional inorganic piezoelectric materials.

## 2 METHODS

### 2.1 DFT CALCULATIONS

All calculations were carried out using the Vienna Ab initio

Simulation Package (VASP) [13], using plane wave basis sets and the projector augmented-wave (PAW) method. Exchange-correlation effects were treated using density functional theory (DFT) via the Perdew, Burke, and Ernzerhof (PBE) implementation of the generalised gradient approximation (GGA). All crystal structures were optimised using conjugate gradient minimisation with a  $4 \times 4 \times 4$  gamma centred k-point grid and plane wave energy cut off of 600 eV.

The stiffness tensor was calculated using a finite differences method, with each atom displaced in each direction by  $\pm 0.01$  Å, with  $8 \times 8 \times 8$  k-point sampling and plane wave energy cut off of 1000 eV to allow the stress tensor to fully converge due to the presence of oxygen and nitrogen atoms [14].

Piezoelectric strain constants and static dielectric tensors were calculated using density functional perturbation theory (DFPT) [15]. Using the piezoelectric charge coefficients  $e_{ij}$  which are calculated directly by VASP, and the elastic stiffness constants  $c_{kj}$ , we can calculate the more useful piezoelectric strain coefficient  $d_{ik}$ , using the relationship:

$$d_{ik} = e_{ij}/c_{kj} \quad (1)$$

To obtain the voltage constant  $g_{ij}$  we divide the piezoelectric strain constant  $d_{ij}$  by the dielectric constant  $\epsilon_{ii}$ , as shown in Equation 2. The voltage constant is an important figure of merit (FoM) for sensor applications, in particular accelerometers and other pressure sensors, and is measured in V m/N.

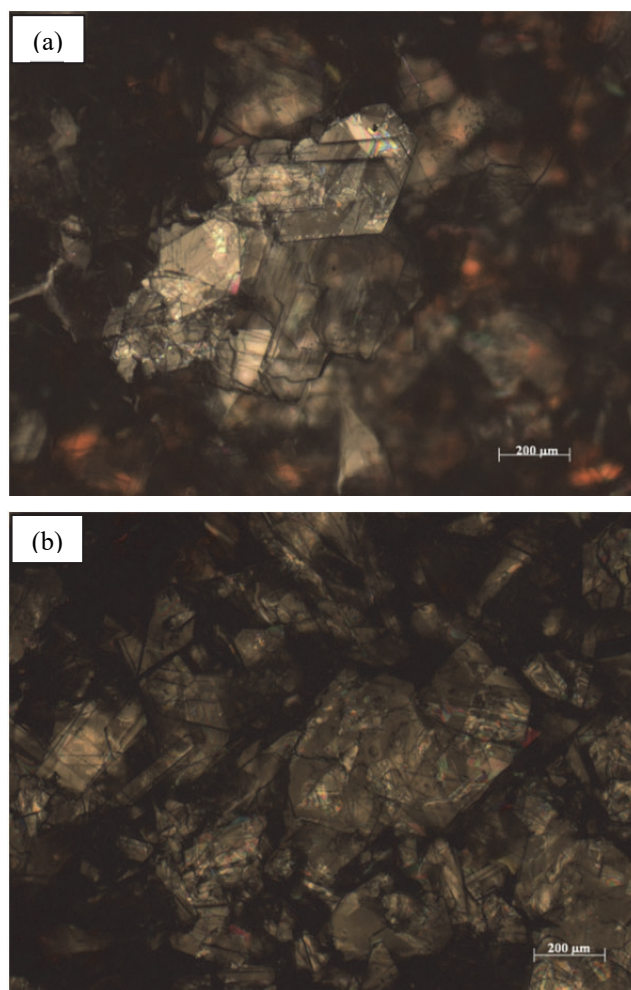
$$g_{ij} = d_{ij}/\epsilon_{ii} \epsilon_0 \quad (2)$$

## 2.2 CRYSTAL GROWTH & CHARACTERISATION

Crystalline films were prepared by dissolving 300 mg of L-leucine in 9ml of de-ionised water. The mixture was heated to 60 °C and stirred until the amino acid powder was completely dissolved. The solution was then drop cast onto clean, dry 20 mm<sup>2</sup> copper substrates. Dense polycrystalline films, of approximately 200 µm thickness and covering the entire substrate, typically appeared within hours and were fully dry within 2-3 days. Optical microscopy images of the crystal films were taken using a Zeiss microscope (Axio Imager.A1m) in dark-field mode. The flake-like crystalline structure typical of L-leucine crystals [16] can be clearly seen in Figure 1.

## 2.3 PIEZOELECTRIC MEASUREMENTS

Longitudinal piezoelectric coefficients of the L-leucine crystal films were measured using a commercial  $d_{33}$  meter (PiezoTest, London, UK), with an accuracy of 0.005 pC/N (pm/V). If the sample under test is piezoelectric, the polarity of the coefficient will change when the sample is inverted. Rather than manually inverting the sample, a simple switch was used to flip the electrode configuration, allowing the L-leucine crystal film to be tested in both inverted and non-inverted states without disturbing the measurement. The copper substrate used for crystal growth acted as the bottom electrode, and a second copper substrate provided the top electrode above the film.



**Figure 1.** Optical microscopy images showing the typical thin flake-like structure of L-leucine amino acid crystals (a) before and (b) after piezoelectric measurements. Though the crystal film is subjected to compressive forces during measurement, the material maintains its crystalline form. Both images are taken in approximately the same area on the same film, but it is not possible to be certain if both images show precisely the same area since, on these length scales, significant displacement of crystals would occur during compressive testing. Scale bar is 200 µm.

## 3 RESULTS

### 3.1 PREDICTED ELASTIC CONSTANTS

Table 1 shows the elastic stiffness constants calculated using density functional theory (DFT) for monoclinic L-leucine crystals. The elastic constants are much smaller than those of typical inorganic piezoelectric materials, highlighting the inherent softness of biological alternatives, which can be very useful in flexible energy harvesting devices and biosensors that require materials that are more malleable than traditional piezoelectric ceramics [17–19].

**Table 1.** Predicted elastic constants for monoclinic L-leucine crystals.

| Elastic Stiffness Constant | GPa |
|----------------------------|-----|
| $c_{11}$                   | 22  |
| $c_{22}$                   | 21  |
| $c_{33}$                   | 20  |
| $c_{44}$                   | 1   |
| $c_{55}$                   | 4   |
| $c_{66}$                   | 8   |

### 3.2 PREDICTED DIELECTRIC CONSTANTS

The dielectric constant or relative permittivity of a material is a measure of the total crystal dipole moment. In the presence of an electric field, bound charges from atomic nuclei and their electrons separate slightly in a dielectric material, inducing a local electric dipole moment. This dipole moment is a product of the electric displacement and the bound charges and is directly linked to the piezoelectric effect of the material. Table 2 shows the predicted values of the relative permittivity tensor which are in the range of 2-3, typical for biological materials and very similar to those calculated for the other monoclinic L-amino acids [6,14].

**Table 2.** Predicted dielectric constants for monoclinic L-leucine crystals.

| Dielectric Constant | (unitless) |
|---------------------|------------|
| $\epsilon_1$        | 2.38       |
| $\epsilon_2$        | 2.43       |
| $\epsilon_3$        | 2.31       |
| $\epsilon_r$        | 2.37       |

The low dielectric constants of amino acid crystals can be very useful. The piezoelectric voltage constant  $g_{ij}$  is inversely proportional to the dielectric constant of the material (Equation 2), meaning that low-dielectric materials with modest piezoelectric strain constants can demonstrate high voltage constants. For example, bacterial phage viruses possess piezoelectric voltage constants of approximately 0.2 V m/N [4], which is comparable to the voltage constants of currently used inorganic materials such as PZT-based ceramics (0.25 V m/N) which have larger piezoelectric strain constants but also higher dielectric constants.

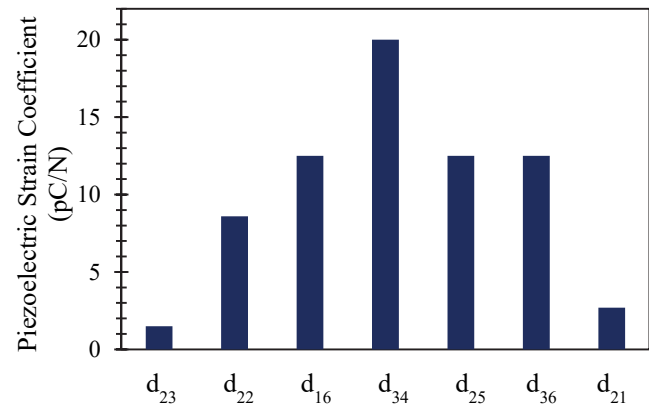
### 3.3 PREDICTED PIEZOELECTRIC CONSTANTS

Quantum mechanical DFT calculations predict L-leucine has up to seven non-zero piezoelectric strain coefficients ( $d_{ij}$ ), varying in magnitude from 1.5 pm/V to 20 pm/V (Figure 2). While the shear piezoelectric coefficients of amino acid crystals tend to be higher, these are often very difficult to exploit in technological applications. As a result, the most useful piezoelectric coefficients tend to be the longitudinal coefficients ( $i = j$ ). In terms of the direct effect, this will represent an induced polarisation in the same direction as the applied stress. L-leucine has one non-zero longitudinal coefficient,  $d_{22}$ , predicted to be 8.6 pm/V. This, along with the transverse coefficients  $d_{23}$  (1.5 pm/V) and  $d_{21}$  (2.7 pm/V), are comparable to the widely used piezoelectric materials quartz [20], zinc oxide [21] and aluminium nitride [22].

L-leucine exhibits corresponding significant piezoelectric voltage constants, ranging from 0.13 to 0.98 V m/N with  $g_{22} = 0.4$  V m/N particularly promising for device applications. This predicted  $g_{22}$  exceeds that of PZT-based ceramics ( $g_{33} = 0.25$  V m/N) and is close to that of bismuth triborate ( $g_{22} = 0.54$  V m/N) [23], indicating that L-leucine could be a useful material for high performance piezoelectric devices.

### 3.4 EXPERIMENTAL PIEZOELECTRIC CONSTANTS

To assess whether crystalline films of L-leucine could demonstrate a useful piezoresponse, we used a commercial



**Figure 2.** Density Functional Theory predicts that monoclinic L-leucine amino acid crystals have a number of modest piezoelectric constants. Each column represents a particular strain constant,  $d_{ij}$  where  $i$  is the direction of the applied stimulus, and  $j$  is the direction of the coupled response

piezometer to carry out longitudinal measurements on films of the amino acid compressed between two copper substrates. Table 3 reports the measured piezoelectric coefficients of three films of L-leucine. The average magnitude of the observed response was +1.57 pC/N in the upright configuration and -1.52 pC/N in the inverted configuration. The consistent sign change upon inversion of the sample is indicative of a genuine piezoelectric response within the films. In all cases, the standard deviations ( $n=5$ ) for each measurement are very low, ranging from 0.01 pC/N to 0.06 pC/N. The observation of the direct piezoelectric effect in films of L-leucine is in agreement with the classical theory of piezoelectricity as monoclinic crystals belonging to space group  $P2_1$  (point group 2) are predicted to exhibit a non-zero longitudinal piezoelectric  $d_{22}$  coefficient. While conventionally the piezometer measures the so-called  $d_{33}$  coefficient, the actual longitudinal coefficient measured by a piezometer may be  $d_{11}$ ,  $d_{22}$  or  $d_{33}$ , depending on the choice of coordinate axis assigned to the laboratory space.

**Table 3.** Measured piezoelectric  $d_{33}$  coefficients for monoclinic L-leucine films.

|                  | Upright $d$ coefficient (pC/N) | Inverted $d$ coefficient (pC/N) |
|------------------|--------------------------------|---------------------------------|
| L-leucine Film 1 | $1.47 \pm 0.03$                | $-1.34 \pm 0.01$                |
| L-leucine Film 2 | $1.36 \pm 0.06$                | $-1.34 \pm 0.05$                |
| L-leucine Film 3 | $1.87 \pm 0.02$                | $-1.89 \pm 0.06$                |

The experimentally measured coefficients are lower than those predicted by DFT calculations on the ideal single crystal. In the experiments, the presence of many single crystals in the crystalline film will attenuate the response. This effect was observed in films of the racemic amino acid DL-alanine which was attributed to an averaging effect caused by the *quasi*-random orientations of individual single crystals within the amino acid film [10]. Optical microscopy (Figure 1) of L-leucine films reveals that a similar effect is at play here, with the varying orientations of the individual single crystals

reducing the potential longitudinal response of the film. The as-grown films do retain a degree of texture and clearly have a preferred orientation, which prevents complete cancellation of the piezoelectric response of each component crystal as would be expected for a completely random packing in the film. Thus, polycrystalline aggregates of amino acids, without poling, can display sizable piezoelectric constants. The piezoelectric response of amino acid films could be maximised with the application of, for example, low temperature corona poling. The response measured here is an order of magnitude larger than that observed in films of the proteinogenic amino acid L-threonine [24] and, to the best of our knowledge, represents the highest piezoelectric strain constant measured in films of a proteinogenic amino acid. The dielectric constants of each film were also determined by noting the capacitance reading during measurement. Values for  $\epsilon_r$  ranged from 2.7 to 3.2, in good agreement with DFT predicted values and in the typical range for amino acid crystals.

## 4 CONCLUSIONS

Amino acids represent an emerging and exciting class of bioelectret materials. Low elastic and dielectric constants, coupled with modest piezoelectric constants, work synergistically to make the proteinogenic amino acid L-leucine a promising candidate material for use in the next generation of flexible, high performance energy harvesting devices. The intrinsic biocompatibility of L-leucine means it is perfectly suited for use in applications requiring non-toxic, biodegradable piezoelectric elements in, for example, self-powered biosensors and medical devices. Beyond single crystals, piezoelectric amino acid films are inexpensive, do not require poling, and can be grown overnight from a simple aqueous solution. The naturally textured nature of L-leucine crystal films also suggests that the modest piezoelectric response could be maximised using poling techniques that do not require very high temperatures.

The work completed to date has paved the way for further investigations involving the use of thin films of L-leucine in both macro- and micro-scale prototype energy harvesting devices. The possible presence of a ferroelectric effect, similar to that seen in  $\beta$ -glycine films grown by evaporative dewetting [25], certainly warrants investigation. Further work is planned to assess the voltage and power output that L-leucine powered electronic circuits may be able to produce under a range of environmental stimuli, including compressive and tensile stresses, as well as temperature cycling to stimulate the likely pyroelectric properties of L-leucine crystals.

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